# Tutorial 4

## Research design exercises

**Task 1.** Form behavioural research questions in usability testing context:

Behavioural research questions on the usability of a product focus on how users interact with a product, the problems they encounter, and the overall user experience. These questions help identify usability issues and areas for improvement. Here is a list of issues around which you can form behavioural research questions on product usability:

- Navigation and Findability
- Error Identification and Recovery
- Task Completion
- Learning Curve
- Mobile Usability
- Form and Data Entry
- Feedback and Help-Seeking Behaviour
- Content Interaction
- Task Priority and Sequencing
- Search Behaviour
- Multimodal Interaction
- Task Switching and Multitasking
- User Preferences
- User Satisfaction and Emotional Responses

Work in pairs. Pick one of the topics above and create a research question.

Make sure that the research question is **clear**, **feasible**, **specific** and **measurable**. This can be achieved if the research question is written so that it outlines various aspects of the study, including:

- ❑ the **population**
- ❑ the **variables** to be studied and
- ❑ the **problem the study addresses**

| Type in here your research question |
| --- |

Share your question to be discussed with the class

Based on the discussion and the comments you received revise your research question.

| Rewrite your research question |
| --- |

**Task 2.** Review the following hypotheses:

Read the following hypotheses and complete the following tasks:

- identify the **dependent and independent variables** in each hypothesis
- identify the **treatments, or conditions** refer to the different techniques, devices, or procedures that we want to compare
- identify the **units** (the objects to which we apply the experiment treatments, in UX context the units are normally human subjects with specific characteristics, such as gender, age, or computing experience)
- identify the **assignment method,** the way in which the experimental units are assigned different treatments

H1. There is no difference in users' reading speed and retention rate when they view news on a desktop computer or a mobile.

H2. There is no difference in the mean selection time across joystick, touch, and gesture.

H3. There is no difference in the technology adoption rate between two speech-based applications with different dialog designs.

H4. There is no difference in the reading skills of children who used educational software for 6 months compared to those who have never used the software.

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**Task 3.** Form a research hypothesis.

Form a hypothesis that will allow you to study the usability of a product. Focus on how users interact with a product, the problems they encounter, and the overall user experience.
Form both the Null and the Alternative hypothesis.

Remember a **hypothesis is a declarative statement about the relationship between two or more variables based on statistical consideration.**

Here is one example:

*RQ: "Among banking customers performing the same transfer task, does using the **mobile app** versus the **web platform** lead to **different task-completion times**?"*

In this hypothesis, you are proposing that there will be a significant difference in task completion times between users of the mobile app and the web-based platform when performing a specific task (transferring funds). This hypothesis sets the stage for a study that will investigate and measure the behavioural differences in usability between the two versions of the product.

**Null Hypothesis ($H_0$):** *"Users of our mobile banking app will not exhibit significantly faster task completion times when transferring funds between accounts compared to users of our web-based banking platform."*
*or*
"Users take the **same average time** to transfer funds on the mobile app and on the web platform."

**Alternative Hypothesis ($H_1$):** *"Users of our mobile banking app will exhibit significantly faster task completion times when transferring funds between accounts compared to users of our web-based banking platform."*
or
*"Users take a **different average time** to transfer funds on the mobile app versus the web platform."*

The mathematical expression of this hypothesis is as follows:

- $H_0$: $\mu_m = \mu_w$ , or $H_0$: $\mu_m - \mu_w = 0$ , or $H_0$: $\mu_m - \mu_w = 0$
- $H_1$: $\mu_m \neq \mu_w$

$\mu_m$ = the mean of mobile banking task completion

$\mu_w$ = the mean of web banking task completion

Work in pairs. Pick one of the topics in task 1 and create a research hypothesis or use the research question you have formed in task 1 and turn this in research hypothesis.

Make sure that the hypothesis is written so that it outlines various aspects of the study states:

- the **dependent and independent variables**
- the **treatments, or conditions**
- the **units**
- and it implies the **assignment method**

| Type in here the Null and the Alternative research hypothesis<br><br>..... |
|---|

Share your question to be discussed with the class

Based on the discussion and the comments you received revise your research question.

| Rewrite your research hypothesis<br><br>..... |
|---|

Try something else, use **Copilot** as a tool to review the correctness of your hypothesis.

| Rewrite your research hypothesis based on **Copilot's** recommendation. Compare this with the reviews you received from your peers.<br><br>..... |
|---|

In your refined hypothesis clearly specify the following:

- the **dependent and independent variables** of the hypothesis
- the **treatments, or conditions**
- the **units**
- the suggested **assignment method you will use and why**

## Task 4. Experiment design

Read the following scenarios. Discuss the type of experiment design (between-group, within-group, or split-plot) that is appropriate for each scenario.

Scenario 1: A study investigates whether people who have attended a security training program generate and use more secure passwords than people who have not received any security training.

Scenario 2: A research team examines the effectiveness of joysticks and trackballs for selecting static targets and moving targets.

Scenario 3: A research team examines whether the gender of an online agent affects the perception of trust for young users, middle-aged users, and older users.

Scenario 4: A research team examines whether virtual teams who use video conferencing are more productive than teams who use phone-based teleconferencing.

Scenario 5: A study examines the effectiveness of three menu structures. The first structure has two levels, with 8 items in the first level and 64 items in the second level. The second structure has three levels, with 4 items in the first level, 16 items in the second level, and 64 items in the third level. The third menu has six levels, with 2 items in the first level and 2 n items in the n the level.

Provide your answer for each scenario:

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